HDOC Day-31 - Solution

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Problem / Objective

Perform menu-driven operations on a 2D array: find the sum of all matrix elements, find the sum of each row and store the results in an array, display the transpose, and exit.

Algorithm

Read rows, columns, and all matrix values. For total sum, use two nested loops. For row sums, reset a sum for each row, accumulate its columns, and store it in a 1D array. For transpose, interchange the traversal order: outer loop over original columns and inner loop over original rows.

C Program

```
#include <stdio.h>
int main(void){
int r,c,i,j,ch;
printf("Rows and columns: ");
scanf("%d%d",&r,&c);
int a[r][c];
printf("Enter matrix elements:\n");
for(i=0;
i<r;
i++)for(j=0;
j<c;
j++)scanf("%d",&a[i][j]);
do{
printf("\n1.Sum all 2.Row sums 3.Transpose 4.Exit\nChoice: ");
scanf("%d",&ch);
switch(ch){
case 1:{
int s=0;
for(i=0;
i<r;
i++)for(j=0;
j<c;
j++)s+=a[i][j];
printf("Sum of all elements = %d\n",s);
break;
}
case 2:{
int sums[r];
for(i=0;
i<r;
i++){
sums[i]=0;
for(j=0;
j<c;
j++)sums[i]+=a[i][j];
printf("Sum of row %d = %d\n",i+1,sums[i]);
}
printf("Row-sum array: ");
for(i=0;
i<r;
i++)printf("%d ",sums[i]);
printf("\n");
break;
}
case 3:printf("Transpose:\n");
for(j=0;
j<c;
j++){
for(i=0;
i<r;
i++)printf("%d ",a[i][j]);
printf("\n");
}
break;
```

```
case 4:break;
default:printf("Invalid choice\n");
}
}
while(ch!=4);
return 0;
}
```

Test Cases / Expected Results

Case	Input	Expected	Status
1	2x3: 1 2 3 / 4 5 6	Total sum = 21	Pass
2	Same matrix	Row sums: 6 and 15; array [6,15]	Pass
3	Same matrix	Transpose rows: 1 4 / 2 5 / 3 6	Pass

Result / Notes

For the sample 2 x 3 matrix, the program reproduces the expected total, row-sum array, and 3 x 2 transpose.